SQL Interview Questions and Answers

Data Science & Machine Learning (Theory)

SQL Basics

1. **What is SQL?**
   SQL (Structured Query Language) is used to create, read, update, and delete data stored in relational databases.

2. **Why is SQL important for Data Scientists?**
   SQL is essential for data extraction, cleaning, aggregation, feature engineering, and preparing datasets for ML models.

3. **What is a relational database?**
   A database that stores data in tables with rows and columns, connected through keys.

4. **What is a table?**
   A structured format consisting of rows (records) and columns (fields) to store data.

5. **What is a schema?**
   A logical structure that defines tables, views, indexes, and relationships in a database.

Keys & Constraints

6. **What is a primary key?**
   A column (or combination) that uniquely identifies each record and cannot be NULL.

7. **What is a foreign key?**
   A column that creates a relationship by referencing the primary key of another table.

8. **What is a candidate key?**
   A column or set of columns that can qualify as a primary key.

9. **What is a composite key?**
   A primary key made from two or more columns.

10. **What are constraints?**
    Rules applied to columns to ensure data integrity, such as NOT NULL, UNIQUE, and CHECK.

Data Filtering & Logic

11. **Difference between WHERE and HAVING?**
    WHERE filters rows before aggregation, HAVING filters results after aggregation.

12. **What is NULL?**
A special value representing missing or unknown data.

13. **Why is NULL handling important in analytics?**
NULL values can distort aggregations and bias ML features if not handled properly.

14. **Difference between AND and OR?**
AND requires all conditions to be true; OR requires at least one condition to be true.

15. **What is a CASE statement?**
Conditional logic in SQL used for feature creation or labeling data.

Joins

16. **What is a JOIN?**
Combines rows from multiple tables using a related column.

17. **INNER JOIN vs LEFT JOIN?**
INNER JOIN returns matched rows only; LEFT JOIN returns all left table rows.

18. **What is RIGHT JOIN?**
Returns all rows from the right table and matching rows from the left.

19. **What is FULL OUTER JOIN?**
Returns all rows from both tables, matched and unmatched.

20. **What is a SELF JOIN?**
Joining a table with itself to analyze hierarchical or relational data.

Aggregation & Analytics

21. **What is aggregation?**
Process of summarizing data using functions like SUM, AVG, and COUNT.

22. **What is GROUP BY?**
Groups rows with the same values to perform aggregation.

23. **COUNT(*) vs COUNT(column)?**
COUNT(*) counts all rows; COUNT(column) ignores NULL values.

24. **What are window functions?**
Functions that perform calculations across related rows without collapsing results.

25. **Why are window functions useful in data science?**
They enable ranking, rolling statistics, and time-based features.

Subqueries & CTEs

26. What is a subquery?
A query nested inside another query to return intermediate results.

27. Subquery vs JOIN?
Subqueries run independently; JOINs combine tables directly.

28. What is a CTE?
A Common Table Expression is a temporary named result set for readability.

29. Advantages of CTEs?
Improves query clarity, reusability, and supports recursion.

30. What is a recursive CTE?
A CTE that references itself to process hierarchical data.

Database Design

31. What is normalization?
The process of organizing data to reduce redundancy and dependency.

32. What are normalization forms?
1NF, 2NF, and 3NF remove different types of anomalies.

33. What is denormalization?
Introducing redundancy to improve read performance.

34. When do data scientists prefer denormalization?
For faster querying and feature extraction.

35. What is referential integrity?
Ensures foreign key values always reference valid primary keys.

Performance & Optimization

36. What is an index?
A performance structure that speeds up data retrieval.

37. How do indexes help queries?
They reduce table scans and improve lookup speed.

38. Can indexes slow performance?
Yes, during INSERT, UPDATE, and DELETE operations.

39. What is an execution plan?
A roadmap showing how the database executes a query.

40. What is partitioning?
Splitting large tables into smaller parts for performance and scalability.

Data Science & ML Context

41. SQL in feature engineering?
To aggregate, transform, and generate behavioral or time-based features.

42. SQL in time-series analysis?
By calculating rolling metrics, lag features, and cumulative values.

43. Identifying outliers using SQL?
Using statistical measures like averages and standard deviation.

44. What is data leakage?
Including future data incorrectly during model training.

45. SQL before model training?
Cleaning data, handling NULLs, labeling outcomes, and aggregations.

Conceptual / Interview Favorites

46. UNION vs UNION ALL?
UNION removes duplicates; UNION ALL keeps them.

47. What is a view?
A virtual table defined by a query.

48. DELETE vs TRUNCATE?
DELETE removes specific rows; TRUNCATE removes all rows.

49. OLTP vs OLAP?
OLTP handles transactions; OLAP handles analytics.

50. SQL vs NoSQL?
SQL uses structured schemas; NoSQL supports flexible schemas.